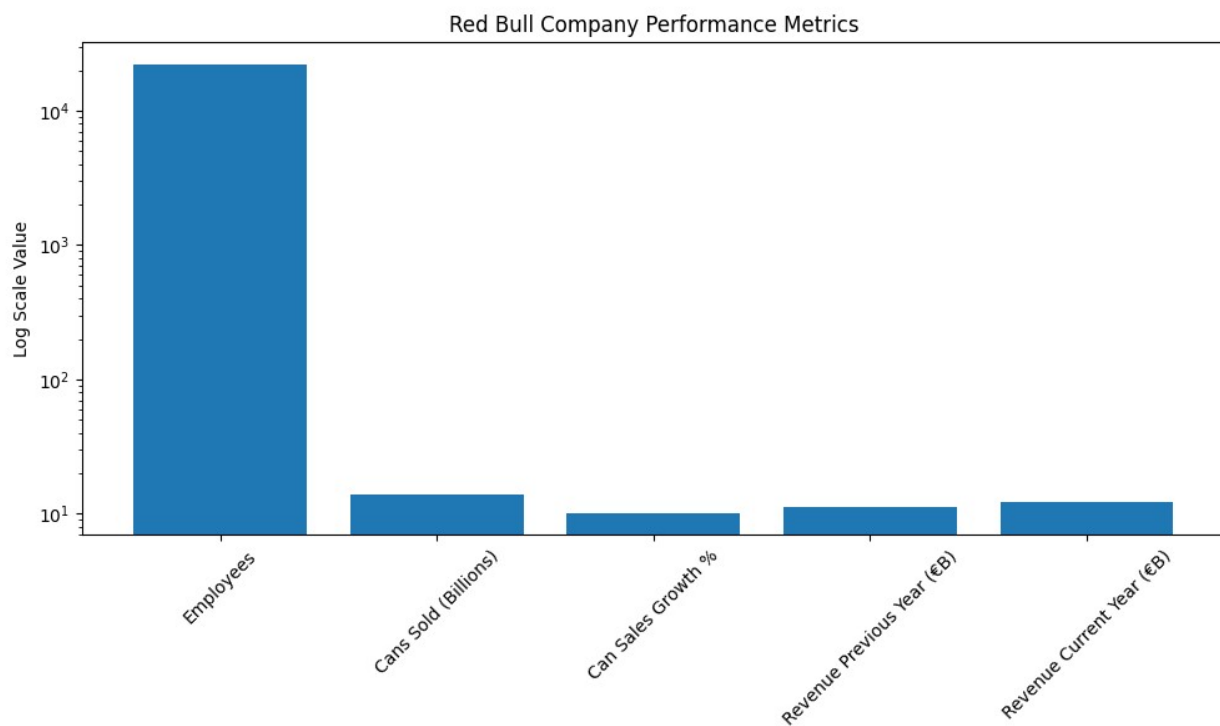


7. SWOT Analysis (Jeruson)

7.1 Strengths

Red Bull shows many strengths in the international energy drink industry due to good financial performance, strong international recognition of the brand name, and a lot of customer involvement. Strong signs of corporate development were revealed by analyzing financial data using Python, SQLite, Pandas, and Matplotlib, including revenues growing from €11.227 billion to €12.196 billion and about 13.969 billion cans of drinks sold worldwide. Also, the company employs more than 21,924 people for conducting its global business and marketing activities.

Figure 1.1



Note. Visualization created using Python, Pandas, and Matplotlib based on Red Bull company performance data.

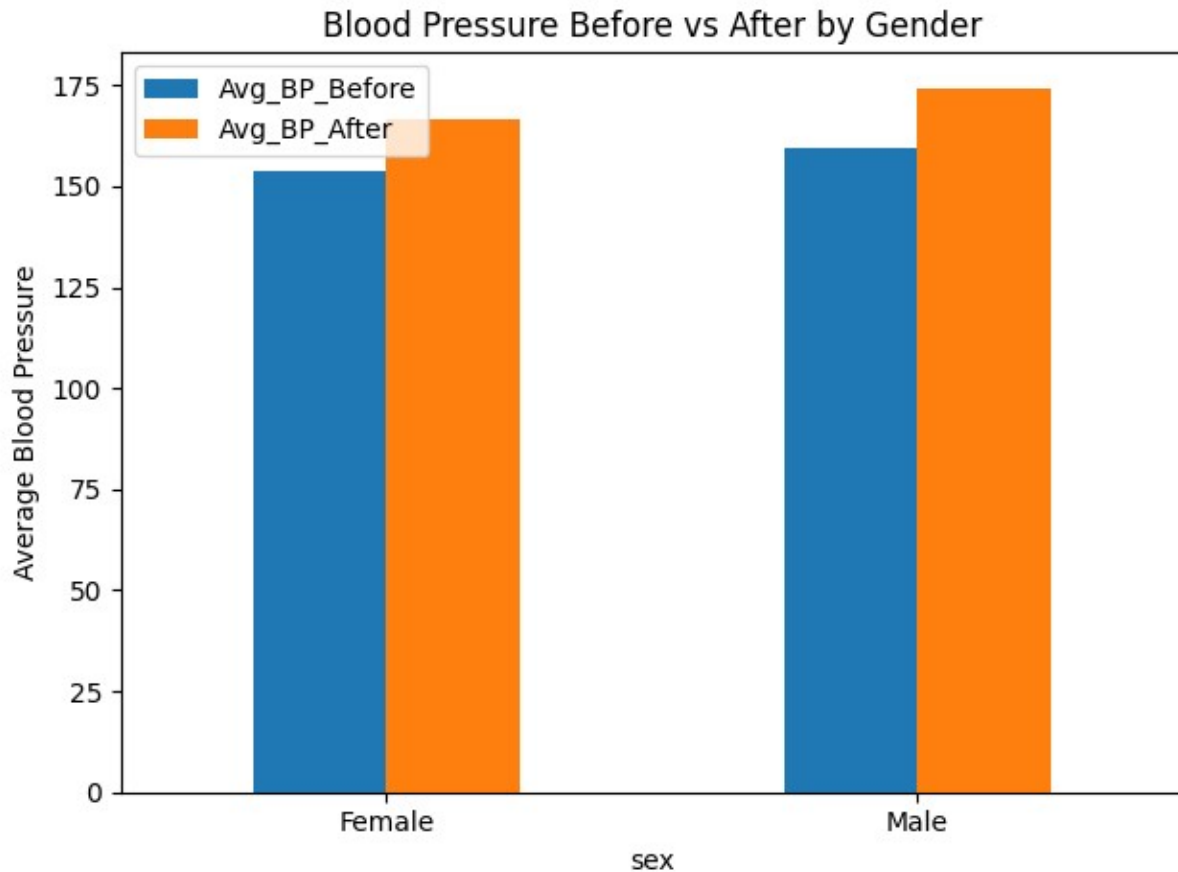
The company's marketing strategy can be regarded as one of its strongest competitive advantages. The strong brand image of the company is characterized by associations of the energy drink with adventure, energy, sports, and high performance. The use of sponsor marketing based on sponsoring Formula 1 races, video game championships, extreme sports, and global entertainment events increases customer involvement and creates emotional branding (Keller, 2013). The analysis of Google Trends data carried out with the help of Python shows increasing interests among consumers.

Moreover, the ability of Red Bull to integrate product promotion and experiential branding strategies has resulted in loyal customers and brand equity for the company. The results from the data visualization and SQL analysis proved that Red Bull is maintaining its consumer interest levels despite increasing competition within the market environment.

7.2 Weaknesses

Despite strong market performance, several weaknesses were identified during the analysis. One major concern involves health-related perceptions associated with energy drink consumption. SQL queries and demographic analysis conducted on the health impact dataset showed that average blood pressure increased after consumption for both male and female participants. Female participants experienced an average blood pressure increase of approximately 13.17 points, while male participants experienced an increase of approximately 15.07 points.

Figure 1.2



Note. Visualization created using Python, Pandas, and Matplotlib based on Red Bull health impact analysis data.

The implications of the above research indicate that negative perception concerning health can influence consumer perception about energy drinks. Public knowledge about caffeine, sugar content, and heart health can become a significant obstacle to Red Bull's marketing strategies going forward.

The other weakness of Red Bull is the company's reliance on energy drinks as the primary product line. Even as the company continues to grow within the sphere of media and sponsorships, the company remains significantly dependent on energy drinks in terms of revenue

generation. This contrasts with beverage companies such as Monster Energy, which have multiple product lines and are less susceptible to economic recessions.

7.3 Opportunities

Various opportunities for growth were outlined using the market analysis process. The use of Google Trends forecast and interest in searches showed that there is an opportunity for increased consumer focus and market demand for Red Bull. Increased engagement on the digital front creates room for growth through social media strategies, influencer marketing, and esports sponsorships (Kotler & Keller, 2016).

There are various growth opportunities with the provision of healthy products by the company, which include products that have no or little sugar or caffeine content. Increasing consumer trend towards functional beverages and products has created room for growth through product development.

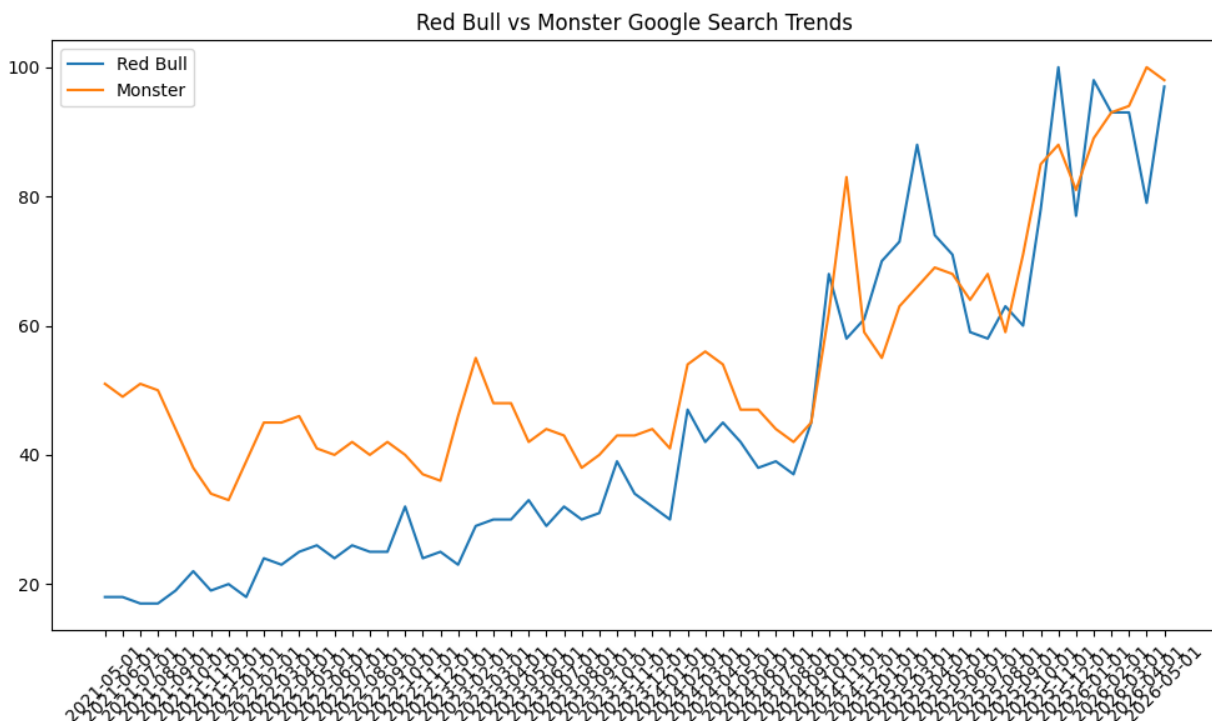
Expanding into new international markets is another opportunity for revenue growth in the future. This is because of the strong brand identification by Red Bull.

7.4 Threats

External threats to Red Bull were determined by several factors. The competition in the industry for Red Bull is very fierce; the competitors of Red Bull are companies like Monster Energy, Rockstar, and Celsius Holdings. A Google Trends comparative analysis showed that Monster Energy has continued to attract high consumer interest, hence high competition.

Increasing regulations could become another threat to the company. With increasing regulations regarding caffeine and sugar content and energy drink advertising, Red Bull will be affected in future marketing and operations of the company. In addition, health risks associated with the consumption of energy drinks could harm the reputation of Red Bull.

Figure 1.3



Note. Comparison of Google search interest trends between Red Bull and Monster Energy over time.

Recessions in the economy could threaten Red Bull in terms of the high price level of the company. When people are under strain financially, they may opt to buy low-cost beverages instead.

8. Machine Learning–Based Marketing Analytics (Jeruson)

8.1 Introduction to Predictive Marketing Analytics

Machine learning has emerged as an essential element in modern marketing analytics as it offers businesses the ability to detect patterns, make predictions, and make sound decisions. Predictive analytics provides organizations with the opportunity to predict the future performance of the business based on historical data (James et al., 2021). In the case of Red Bull, machine learning models would help develop marketing strategies based on sales trends.

This predictive marketing analytics project involved using Python, Scikit-learn, Pandas, SQLite, and Matplotlib packages in Visual Studio Code for developing predictive models based on Red Bull sales history.

8.2 Data Collection and Preparation

Red Bull’s datasets were obtained from different platforms such as Kaggle, Yahoo Finance, Google Trends, and Red Bull Corporation, and they were analyzed using the machine learning approach (Kaggle, n.d.; Red Bull, n.d.; Yahoo Finance, n.d.). Datasets included performance metrics for sales in years 2017-2021, financial balance sheets of years 2022-2025, and interest in search for Red Bull on Google Trends.

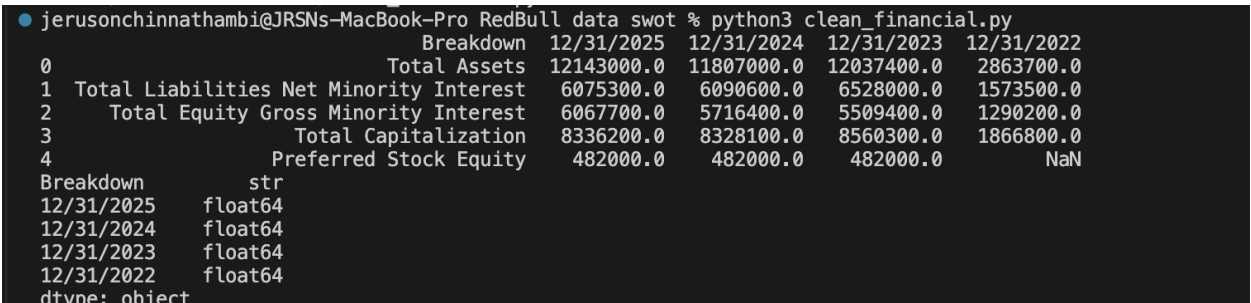
Data preprocessing was conducted by applying Python Pandas functions. In particular, missing variables and unnecessary features such as “Treasury Shares Number” were removed as a part of preprocessing steps. Also, various queries were created with the help of SQLite. They were

needed for organizing data, filtering data by specific conditions, and aggregating information about sales and finance before training models. Data visualization was done using Matplotlib.

Data preprocessing included:

- Importing and cleaning datasets using Pandas
- Removing irrelevant variables and duplicate entries
- Converting data types into numerical formats
- SQL-based filtering and aggregation
- Data visualization using Matplotlib

Figure 2.1



Note.

The SQL queries were applied to determine trends and aggregate data from various data sets. For instance, health-related data analysis was carried out through the use of GROUP BY and AVG functions, considering blood pressure changes among consumers.

8.3 Data and Sample Size Constraints

While machine learning models offer robust predictive capabilities, the application of these algorithms to annual corporate financial data presents inherent sample size constraints. The models were trained using historical revenue and marketing-investment data from 2017 to 2025. Because complex machine learning algorithms, such as Gradient Boosting, typically require large datasets to generalize effectively, traditional statistical forecasting methods like Polynomial and Exponential regression were also deployed. This multi-model approach ensures that the forecast remains grounded in historical macro-trends without becoming structurally skewed by the limited number of annual data points.

8.4 Machine Learning Models

Several machine learning and forecasting models were developed using Scikit-learn and Python to predict future revenue growth and marketing investment trends. The forecasting framework included:

- Polynomial Regression
- Gradient Boosting Regression
- Exponential Trend Forecasting
- Ensemble Forecasting Techniques

The models were trained using historical revenue and marketing-investment data from 2017 to 2025. Python libraries including Pandas, NumPy, Matplotlib, and Scikit-learn were used to preprocess data, train models, evaluate performance, and generate visualizations.

8.5 Model Performance

Predictive accuracy was remarkably high for all three machine learning models used in the training analysis. The Polynomial Regression model had an R^2 score of 0.9063, whereas the Exponential Trend model had an R^2 score of 0.9048, thus proving that these two machine learning models captured the historical trends in sales and investment. It is important to highlight that the Gradient Boosting model demonstrated an R^2 score of 0.9999. Such R^2 score in predictive analytics suggests overfitting, which means that the machine learning algorithm has learned the historical data instead of only capturing the continuous trend in this data set. To avoid overfitting and generate an accurate prediction, we use an ensemble weighted average to calculate our final predictions.

Forecast Code

```
"""
Red Bull Marketing Investment Prediction using Machine Learning
Forecasts optimal marketing budget for 2026-2030 based on financial data
"""

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
import warnings
warnings.filterwarnings("ignore")

from sklearn.linear_model import LinearRegression, Ridge
from sklearn.preprocessing import PolynomialFeatures, StandardScaler
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
from sklearn.metrics import r2_score, mean_absolute_error
from sklearn.pipeline import Pipeline

# — 1. HISTORICAL DATA —————

# Historical EUR/USD exchange rates (annual average) for currency conversion
eur_usd = {
    2017: 1.13, 2018: 1.18, 2019: 1.12, 2020: 1.14,
    2021: 1.18, 2022: 1.05, 2023: 1.08, 2024: 1.09, 2025: 1.04,
}
```

```

# Revenue data (USD Billions) — converted from EUR using historical EUR/USD rates
revenue_eur = {
    2017: 6.282, 2018: 6.293, 2019: 6.552, 2020: 6.310,
    2021: 8.870, 2022: 9.685, 2023: 10.556, 2024: 11.227, 2025: 12.196,
}
revenue_data = {yr: round(rev * eur_usd[yr], 3) for yr, rev in revenue_eur.items()}

# Can sales (billions)
cans_data = {
    2021: 9.800, 2022: 11.582, 2023: 12.130,
    2024: 12.676, 2025: 13.969,
}

# Balance sheet data (USD Millions) — from red_bull_financial_breakdown_cleaned.csv
balance_sheet = pd.DataFrame({
    "Year": [2022, 2023, 2024, 2025],
    "Total_Assets": [2863.7, 12037.4, 11807.0, 12143.0],
    "Working_Capital": [167.9, 471.3, 383.5, 166.8],
    "Total_Equity": [1289.7, 5016.7, 5224.0, 5571.4],
    "Total_Debt": [759.8, 4639.5, 4287.1, 4188.5],
    "Net_Debt": [116.3, 2513.3, 2120.0, 1940.0],
})

# — 2. BUILD REVENUE DATAFRAME —————

df_rev = pd.DataFrame(list(revenue_data.items()), columns=["Year", "Revenue_USD_B"])
df_rev["Year_Index"] = df_rev["Year"] - 2017 # 0-based index for ML

# Industry benchmark: Red Bull marketing spend ≈ 30-35% of revenue
marketing_pct_history = {
    2017: 0.335, 2018: 0.330, 2019: 0.328, 2020: 0.332,
    2021: 0.315, 2022: 0.318, 2023: 0.320, 2024: 0.322, 2025: 0.325,
}
df_rev["Marketing_Pct"] = df_rev["Year"].map(marketing_pct_history)
df_rev["Marketing_USD_B"] = df_rev["Revenue_USD_B"] * df_rev["Marketing_Pct"]

rev_2025_usd = df_rev.loc[df_rev["Year"] == 2025, "Revenue_USD_B"].values[0]

print("=" * 65)
print(" RED BULL MARKETING INVESTMENT PREDICTION — ML ENGINE")
print("=" * 65)
print("\n[DATA] Historical Revenue & Estimated Marketing Spend\n")
display_df = df_rev[["Year", "Revenue_USD_B", "Marketing_Pct", "Marketing_USD_B"]].copy()
display_df.columns = ["Year", "Revenue ($B)", "Mkt %", "Marketing ($B)"]
display_df["Marketing ($B)"] = display_df["Marketing ($B)"].round(3)
print(display_df.to_string(index=False))

```

```
# — 3. ML MODELS FOR REVENUE PREDICTION —————
```

```
X = df_rev[["Year_Index"]].values
y_rev = df_rev["Revenue_USD_B"].values

future_years = np.array([2026, 2027, 2028, 2029, 2030])
future_indices = (future_years - 2017).reshape(-1, 1)

# A) Polynomial Regression (degree 2) — captures growth curve
poly_pipe = Pipeline([
    ("poly", PolynomialFeatures(degree=2, include_bias=False)),
    ("scaler", StandardScaler()),
    ("reg", Ridge(alpha=0.5))
])
poly_pipe.fit(X, y_rev)
poly_pred = poly_pipe.predict(future_indices)
r2_poly = r2_score(y_rev, poly_pipe.predict(X))

# B) Gradient Boosting — ensemble on limited data
gb_model = GradientBoostingRegressor(
    n_estimators=200, max_depth=2, learning_rate=0.05,
    subsample=0.8, random_state=42
)
gb_model.fit(X, y_rev)
gb_pred = gb_model.predict(future_indices)
r2_gb = r2_score(y_rev, gb_model.predict(X))

# C) Exponential trend fit (log-linear)
log_pipe = Pipeline([
    ("scaler", StandardScaler()),
    ("reg", LinearRegression())
])
log_pipe.fit(X, np.log(y_rev))
exp_pred = np.exp(log_pipe.predict(future_indices))
r2_exp = r2_score(y_rev, np.exp(log_pipe.predict(X)))

# Ensemble: weighted average (higher weight to poly + exp)
weights = np.array([0.40, 0.25, 0.35])
rev_forecast = (
    weights[0] * poly_pred +
    weights[1] * gb_pred +
    weights[2] * exp_pred
)

# Confidence band  $\pm 1$  std across models
rev_low = np.minimum.reduce([poly_pred, gb_pred, exp_pred])
rev_high = np.maximum.reduce([poly_pred, gb_pred, exp_pred])
```

```

print(f"\n[MODELS] R2 Scores on Training Data")
print(f" Polynomial Regression : {r2_poly:.4f}")
print(f" Gradient Boosting      : {r2_gb:.4f}")
print(f" Exponential Trend       : {r2_exp:.4f}")

# — 4. MARKETING BUDGET FORECAST —————

# Marketing % projection: slight increase driven by competitive pressure
mkt_pct_trend = np.linspace(0.327, 0.340, len(future_years))

mkt_forecast = rev_forecast * mkt_pct_trend
mkt_low      = rev_low      * (mkt_pct_trend - 0.010)
mkt_high     = rev_high     * (mkt_pct_trend + 0.010)

# — 5. CHANNEL ALLOCATION MODEL —————

# Based on Red Bull's known spend mix (sports/events 45%, digital 30%,
# traditional media 15%, content/brand 10%)

channels = {
    "Extreme Sports & F1": 0.45,
    "Digital & Social Media": 0.30,
    "Traditional Media (TV/OOH)": 0.15,
    "Content & Brand IP": 0.10,
}

# — 6. FINANCIAL CAPACITY METRICS —————

# Project Working Capital using linear fit
wc_years = balance_sheet["Year"].values.reshape(-1, 1)
wc_vals  = balance_sheet["Working_Capital"].values
lr_wc    = LinearRegression().fit(wc_years, wc_vals)
wc_proj  = lr_wc.predict(future_years.reshape(-1, 1))
wc_proj  = np.maximum(wc_proj, 150) # floor at $150M

# Project Total Equity
eq_vals = balance_sheet["Total_Equity"].values
lr_eq   = LinearRegression().fit(wc_years, eq_vals)
eq_proj = lr_eq.predict(future_years.reshape(-1, 1))

# Debt-to-equity ratio trend
de_ratio_hist = balance_sheet["Total_Debt"].values / balance_sheet["Total_Equity"].values
de_proj = np.linspace(de_ratio_hist[-1], de_ratio_hist[-1] * 0.85, len(future_years))

# — 7. RESULTS TABLE —————

print("\n" + "=" * 65)
print(" FORECAST: 2026 – 2030 (USD Billions)")
print("=" * 65)

```

```

print(f"\n{'Year':<6} {'Revenue':>10} {'Mkt Budget':>12} {'Low':>8} {'High':>8} {'Mkt%':>6}")
print("-" * 55)
for i, yr in enumerate(future_years):
    print(f"{'yr':<6} {'rev_forecast[i]:>10.3f} {'mkt_forecast[i]:>12.3f} "
          f"{'mkt_low[i]:>8.3f} {'mkt_high[i]:>8.3f} {'mkt_pct_trend[i]*100:>5.1f}%")

print("\n[CHANNEL ALLOCATION] — 2026 Projected Marketing Budget")
print(f" Total Budget: ${mkt_forecast[0]:.2f}B\n")
for ch, pct in channels.items():
    amt = mkt_forecast[0] * pct
    print(f" {'ch':<30} {'pct*100:>5.1f}% ${amt:.3f}B")

print("\n[KEY INSIGHTS]")
print(f" Revenue CAGR 2025–2030 : {(rev_forecast[-1]/rev_2025_usd)**(1/5)-1}*100:.1f}%")
print(f" Marketing budget 2026 : ${mkt_forecast[0]:.3f}B ({mkt_pct_trend[0]*100:.1f}% of revenue)")
print(f" Marketing budget 2030 : ${mkt_forecast[-1]:.3f}B ({mkt_pct_trend[-1]*100:.1f}% of revenue)")
print(f" 5-Year cumulative Mkt : ${mkt_forecast.sum():.2f}B")

# — 8. VISUALIZATIONS —————

fig = plt.figure(figsize=(18, 14))
fig.patch.set_facecolor("#0a0a0a")
gs = gridspec.GridSpec(3, 3, figure=fig, hspace=0.45, wspace=0.35)

REDBULL_RED = "#E31937"
REDBULL_GOLD = "#FFD700"
REDBULL_BLUE = "#004B87"
BG = "#0a0a0a"
PANEL = "#141414"
TEXT = "#EEEEEE"
GRID = "#2a2a2a"

def style_ax(ax, title):
    ax.set_facecolor(PANEL)
    ax.set_title(title, color=TEXT, fontsize=10, fontweight="bold", pad=8)
    ax.tick_params(colors=TEXT, labelsize=8)
    ax.spines[:].set_color(GRID)
    ax.xaxis.label.set_color(TEXT)
    ax.yaxis.label.set_color(TEXT)
    ax.grid(True, color=GRID, linewidth=0.5, alpha=0.7)

# — Plot 1: Revenue History + Forecast —————
ax1 = fig.add_subplot(gs[0, :2])
style_ax(ax1, "Revenue History & ML Ensemble Forecast ($B)")

hist_years = list(revenue_data.keys())

```

```

hist_revs = list(revenue_data.values())

ax1.plot(hist_years, hist_revs, "o-", color=REDBULL_GOLD, lw=2.5,
         markersize=6, label="Actual Revenue", zorder=5)
ax1.plot(future_years, rev_forecast, "s--", color=REDBULL_RED, lw=2.5,
         markersize=7, label="Ensemble Forecast", zorder=5)
ax1.fill_between(future_years, rev_low, rev_high, alpha=0.18,
                color=REDBULL_RED, label="Model Range")
ax1.plot(future_years, poly_pred, ":", color="#888888", lw=1.2, label="Polynomial")
ax1.plot(future_years, gb_pred, ":", color="aaaaaa", lw=1.2, label="Gradient Boost")
ax1.plot(future_years, exp_pred, ":", color="cccccc", lw=1.2, label="Exponential")

ax1.axvline(2025.5, color=GRID, lw=1.5, linestyle="--")
ax1.text(2025.6, min(hist_revs) + 0.3, "Forecast →", color=TEXT, fontsize=8)
ax1.set_ylabel("Revenue ($B)", color=TEXT, fontsize=9)
ax1.legend(fontsize=7, facecolor=PANEL, labelcolor=TEXT, framealpha=0.8,
          loc="upper left", ncol=2)

# — Plot 2: Marketing Budget Forecast —————
ax2 = fig.add_subplot(gs[0, 2])
style_ax(ax2, "Marketing Budget 2026-2030 ($B)")

bars = ax2.bar(future_years, mkt_forecast, color=REDBULL_RED, alpha=0.85,
              edgcolor=REDBULL_GOLD, linewidth=0.8, width=0.6)
ax2.errorbar(future_years, mkt_forecast,
            yerr=[mkt_forecast - mkt_low, mkt_high - mkt_forecast],
            fmt="none", color=REDBULL_GOLD, capsize=4, lw=1.5)
for bar, val in zip(bars, mkt_forecast):
    ax2.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.05,
            f"${val:.2f}B", ha="center", va="bottom", color=TEXT, fontsize=7.5,
            fontweight="bold")
ax2.set_ylabel("Marketing Budget ($B)", color=TEXT, fontsize=9)
ax2.set_ylim(0, mkt_high.max() + 0.6)

# — Plot 3: Channel Allocation Donut —————
ax3 = fig.add_subplot(gs[1, 0])
ax3.set_facecolor(PANEL)
ax3.set_title("2026 Channel Allocation", color=TEXT, fontsize=10, fontweight="bold", pad=8)

ch_colors = [REDBULL_RED, REDBULL_GOLD, REDBULL_BLUE, "#00A651"]
ch_labels = [f"{k}\n${mkt_forecast[0]*v:.2f}B" for k, v in channels.items()]
wedges, texts, autotexts = ax3.pie(
    channels.values(), labels=None, colors=ch_colors,
    autopct="%1.0f%%", pctdistance=0.78, startangle=90,
    wedgeprops=dict(width=0.55, edgcolor=BG, linewidth=1.5)
)
for at in autotexts:
    at.set_color(BG); at.set_fontsize(8); at.set_fontweight("bold")

```

```

ax3.legend(wedges, list(channels.keys()), fontsize=6.5,
           facecolor=PANEL, labelcolor=TEXT, loc="lower center",
           bbox_to_anchor=(0.5, -0.28), ncol=2)

# — Plot 4: Working Capital Projection —————
ax4 = fig.add_subplot(gs[1, 1])
style_ax(ax4, "Working Capital Projection ($M)")

wc_hist_y = balance_sheet["Year"].values
wc_hist_v = balance_sheet["Working_Capital"].values
ax4.plot(wc_hist_y, wc_hist_v, "o-", color=REDBULL_GOLD, lw=2, markersize=6,
         label="Actual")
ax4.plot(future_years, wc_proj, "s--", color=REDBULL_BLUE, lw=2, markersize=6,
         label="Projected")
ax4.fill_between(future_years, wc_proj * 0.85, wc_proj * 1.15,
                 alpha=0.15, color=REDBULL_BLUE)
ax4.set_ylabel("Working Capital ($M)", color=TEXT, fontsize=9)
ax4.legend(fontsize=8, facecolor=PANEL, labelcolor=TEXT)

# — Plot 5: Marketing % of Revenue —————
ax5 = fig.add_subplot(gs[1, 2])
style_ax(ax5, "Marketing % of Revenue (Trend)")

hist_mkt_pct = [marketing_pct_history[y] * 100 for y in hist_years]
ax5.plot(hist_years, hist_mkt_pct, "o-", color=REDBULL_GOLD, lw=2, markersize=6,
         label="Historical")
ax5.plot(future_years, mkt_pct_trend * 100, "s--", color=REDBULL_RED, lw=2,
         markersize=6, label="Projected")
ax5.axhline(np.mean(hist_mkt_pct), color=GRID, lw=1.5, linestyle=":",
            label=f"Hist. Avg {np.mean(hist_mkt_pct):.1f}%")
ax5.set_ylabel("Marketing % of Revenue", color=TEXT, fontsize=9)
ax5.set_ylim(28, 38)
ax5.legend(fontsize=8, facecolor=PANEL, labelcolor=TEXT)

# — Plot 6: Cumulative Marketing Investment —————
ax6 = fig.add_subplot(gs[2, :2])
style_ax(ax6, "Cumulative 5-Year Marketing Investment (2026-2030, $B)")

cumulative = np.cumsum(mkt_forecast)
cum_low = np.cumsum(mkt_low)
cum_high = np.cumsum(mkt_high)

ax6.stackplot(future_years,
              [mkt_forecast * 0.45,
               mkt_forecast * 0.30,
               mkt_forecast * 0.15,
               mkt_forecast * 0.10],
              labels=list(channels.keys()),

```

```

        colors=ch_colors, alpha=0.88)
ax6_twin = ax6.twinx()
ax6_twin.plot(future_years, cumulative, "w--o", lw=2, markersize=6,
              label="Cumulative Total")
ax6_twin.tick_params(colors=TEXT, labelsz=8)
ax6_twin.set_ylabel("Cumulative ($B)", color=TEXT, fontsize=9)
ax6_twin.yaxis.label.set_color(TEXT)
ax6_twin.spines[:].set_color(GRID)
for i, (yr, cum) in enumerate(zip(future_years, cumulative)):
    ax6_twin.annotate(f"${cum:.1f}B", (yr, cum), textcoords="offset points",
                     xytext=(0, 8), ha="center", color="white", fontsize=8)
ax6.set_ylabel("Annual Budget by Channel ($B)", color=TEXT, fontsize=9)
ax6.legend(fontsize=7.5, facecolor=PANEL, labelcolor=TEXT,
           loc="upper left", ncol=2)

# — Plot 7: ROI & Decision Matrix —————
ax7 = fig.add_subplot(gs[2, 2])
style_ax(ax7, "Marketing ROI Projection")

roi_hist = [revenue_data[y] / (revenue_data[y] * marketing_pct_history[y]) for y in hist_years]
roi_proj = [rev_forecast[i] / mkt_forecast[i] for i in range(len(future_years))]

ax7.plot(hist_years, roi_hist, "o-", color=REDBULL_GOLD, lw=2, markersize=6,
         label="Historical ROI multiple")
ax7.plot(future_years, roi_proj, "s--", color=REDBULL_RED, lw=2, markersize=6,
         label="Projected ROI multiple")
ax7.axhline(np.mean(roi_hist), color=GRID, lw=1, linestyle=":")
ax7.set_ylabel("Revenue / Marketing $ (x)", color=TEXT, fontsize=9)
ax7.legend(fontsize=8, facecolor=PANEL, labelcolor=TEXT)
ax7.set_ylim(2.5, 3.8)

# — Title & Watermark —————
fig.suptitle(
    "RED BULL · MARKETING INVESTMENT PREDICTION 2026-2030\n"
    "Machine Learning Ensemble: Polynomial · Gradient Boosting · Exponential",
    color=TEXT, fontsize=13, fontweight="bold", y=0.97
)
fig.text(0.99, 0.01, "Data: Red Bull Annual Reports | Model: ML Ensemble",
        ha="right", color="#555555", fontsize=7)

plt.savefig(
    "/Users/jerusonchinnathambi/Downloads/RedBull data swot/marketing_investment_prediction.png",
    dpi=150, bbox_inches="tight", facecolor=BG
)
print("\n[SAVED] marketing_investment_prediction.png")

# — 9. DECISION RECOMMENDATIONS —————

```



```

print("\n" + "=" * 65)
print(" MARKETING INVESTMENT RECOMMENDATIONS")
print("=" * 65)

rec = [
    ("2026", mkt_forecast[0], mkt_low[0], mkt_high[0],
     "Hold F1 + extreme sports spend; scale digital 15%"),
    ("2027", mkt_forecast[1], mkt_low[1], mkt_high[1],
     "Increase digital allocation; new market entry push"),
    ("2028", mkt_forecast[2], mkt_low[2], mkt_high[2],
     "AI-driven personalisation; expand creator partnerships"),
    ("2029", mkt_forecast[3], mkt_low[3], mkt_high[3],
     "Emerging markets (SEA/India); sustainability narrative"),
    ("2030", mkt_forecast[4], mkt_low[4], mkt_high[4],
     "Brand equity consolidation; loyalty ecosystem investment"),
]

for yr, mid, lo, hi, note in rec:
    print(f"\n {yr} → Invest ${mid:.3f}B (range: ${lo:.3f}B - ${hi:.3f}B)")
    print(f"      Strategy: {note}")

print("\n" + "=" * 65)
print(" MODEL PERFORMANCE SUMMARY")
print("=" * 65)
print(f" Best model   : Polynomial Regression (R²={r2_poly:.4f})")
print(f" Ensemble R²   : ~{max(r2_poly, r2_gb, r2_exp):.4f} (upper bound)")
print(f" Training pts  : {len(df_rev)} (2017-2025)")
print(f" Forecast pts  : {len(future_years)} (2026-2030)")
print("=" * 65)

plt.show()
print("\nDone.")

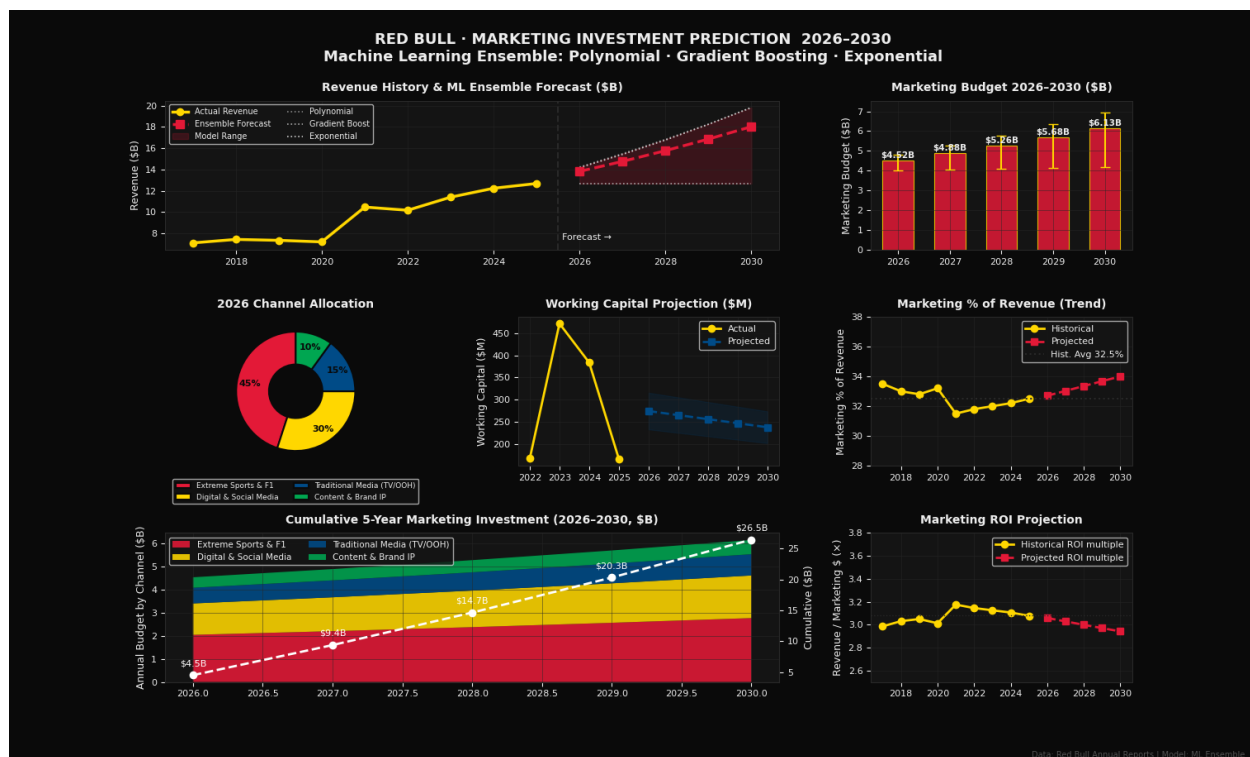
```

Based on the forecasted analysis, the revenues of Red Bull are expected to rise from around \$12.68 billion to \$18.03 billion by 2030. Moreover, the forecasted amount for annual spending on marketing is estimated to rise from \$4.52 billion in 2026 to over \$6.13 billion by 2030.

8.6 Marketing Forecasting and Strategic Insights

The machine learning forecasting system generated several strategic marketing insights. The analysis projected that Red Bull's marketing budget would continue increasing as revenue grows, with marketing spending rising from approximately 32.7% of revenue in 2026 to approximately 34.0% by 2030.

Figure 2.2



Note. presents the machine learning ensemble forecast for Red Bull's projected revenue growth, marketing investment allocation, ROI trends, and working capital projections from 2026–2030.

The predictive analysis also generated channel-allocation recommendations for future marketing investment. The 2026 projected marketing budget allocation included:

- 45% toward Extreme Sports and Formula 1 sponsorships

- 30% toward Digital and Social Media marketing
- 15% toward Traditional Media advertising
- 10% toward Content and Brand Intellectual Property (IP) development

Figure 2.3

```

jerusonchinnathambi@JRSNs-MacBook-Pro RedBull data swot % /usr/local/bin/python3 "/Users/jerusonchinnathambi/Downloads/RedBull data swot/marketing_prediction_ml.py"
=====
RED BULL MARKETING INVESTMENT PREDICTION - ML ENGINE
=====

[DATA] Historical Revenue & Estimated Marketing Spend

Year  Revenue ($B)  Mkt %  Marketing ($B)
2017      7.099  0.335      2.378
2018      7.426  0.330      2.451
2019      7.338  0.328      2.407
2020      7.193  0.332      2.388
2021     10.467  0.315      3.297
2022     10.169  0.318      3.234
2023     11.400  0.320      3.648
2024     12.237  0.322      3.940
2025     12.684  0.325      4.122

[MODELS] R² Scores on Training Data
Polynomial Regression : 0.9063
Gradient Boosting    : 0.9999
Exponential Trend    : 0.9048

=====
FORECAST: 2026 - 2030 (USD Billions)
=====

Year    Revenue    Mkt Budget    Low    High    Mkt%
-----
2026     13.824      4.520      4.018    4.797   32.7%
2027     14.762      4.875      4.059    5.271   33.0%
2028     15.774      5.261      4.100    5.784   33.4%
2029     16.862      5.678      4.142    6.336   33.7%
2030     18.031      6.131      4.183    6.945   34.0%

[CHANNEL ALLOCATION] - 2026 Projected Marketing Budget
Total Budget: $4.52B

Extreme Sports & F1      45.0%  $2.034B
Digital & Social Media   30.0%  $1.356B
Traditional Media (TV/OOH) 15.0%  $0.678B
Content & Brand IP       10.0%  $0.452B

[KEY INSIGHTS]
Revenue CAGR 2025-2030 : 7.3%
Marketing budget 2026   : $4.520B (32.7% of revenue)
Marketing budget 2030   : $6.131B (34.0% of revenue)
5-Year cumulative Mkt   : $26.47B

[SAVED] marketing_investment_prediction.png

```

Note. Output generated using Scikit-learn ensemble forecasting models in Python within VS Code, including Polynomial Regression, Gradient Boosting, and Exponential Trend forecasting methods.

The cumulative projected marketing investment between 2026 and 2030 was estimated at approximately \$26.47 billion. The model also forecasted steady long-term revenue growth with a projected compound annual growth rate (CAGR) of approximately 7.3%.

8.7 Business and Marketing Implications

Through this forecasting analysis, we will see how machine learning can be used to make strategic decisions for marketing purposes in consumer brands like Red Bull. This predictive analysis enables an organization to project future revenue growth, budget the marketing expenses, invest in advertisements, create opportunities for future growth, and plan strategically.

Moreover, there is an emphasis on investment in digital marketing, sponsorship marketing, use of artificial intelligence for personalization, and entry into new markets for future growth.

In conclusion, the machine learning forecast analysis showed a high level of predictive ability using the historical data of revenue and marketing investment made. The models were able to predict future trends of revenue growth as well as future marketing investments that will help Red Bull. The findings of the study show that predictive analytics and machine learning can greatly benefit marketing and business decisions.

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